

Library Management System

Project Description:-

The Library Management System is a Python-based application developed using Object-Oriented Programming (OOP) principles to automate the basic operations of a library. The system allows librarians to manage books and members efficiently by providing features such as adding books, registering members, borrowing books, returning books, searching for books, and tracking the availability of library resources. The project is designed around three main classes: Book, Member, and Library. The Book class stores information about each book, including its ID, title, author, and availability status. The Member class maintains member details and keeps track of the books borrowed by each member.

What I Learned:-

- ✓ Understanding the fundamentals of **Object-Oriented Programming (OOP)**.
- ✓ Creating and using **classes and objects** in Python.
- ✓ Applying **encapsulation** by combining data and methods within classes.
- ✓ Managing relationships between multiple classes such as **Book, Member, and Library**.
- ✓ Using **lists** to store and manage collections of objects.
- ✓ Implementing **functions and methods** to perform specific operations.

How to Run This Program:-

- Install Python 3 on your computer.
- Open a code editor such as IDLE, VS Code, or PyCharm.
- Save the code as `library_management_system.py`.
- Open a terminal or command prompt in the file's folder.
- Run the program using:

```
{python library_management_system.py}
```
- Follow the on-screen menu to add books, register members, borrow or return books, search, and view the library summary.

Features of the Program:-

- Object-Oriented Programming with Book, Member, and Library classes.
- Add new books to the library. Register new members.
- Borrow and return books.
- Track book availability.
- Search books by title.
- Display all books and members.

Challenges & Solutions:-

1. Challenge 1: Managing relationships between books and members

Solution: Stored borrowed Book objects inside each Member object.

2. Challenge 2: Preventing duplicate borrowing

Solution: Used an available attribute in the Book class to track availability.

3. Challenge 3: Finding books and members efficiently

Solution: Implemented find_book() and find_member() helper methods.

4. Challenge 4: Handling invalid user input

Solution: Used try/except blocks to catch incorrect input types.

5. Challenge 4: Keeping library records organized

Solution: Centralized all operations inside the Library class

OUTPUT:-

===== LIBRARY MANAGEMENT SYSTEM =====

1. Add Book
2. Register Member
3. Borrow Book
4. Return Book
5. Display Books
6. Display Members
7. Search Book
8. Library Summary
9. Exit

Enter your choice:

SCREENSHOTS:

```
===== LIBRARY MANAGEMENT SYSTEM =====
```

1. Add Book
2. Register Member
3. Borrow Book
4. Return Book
5. Display Books
6. Display Members
7. Search Book
8. Library Summary
9. Exit

Enter your choice: 1

Book ID: 52648

Book Title: Python

Author: D.Arun Kumar

Book added successfully.

```
===== LIBRARY MANAGEMENT SYSTEM =====
```

1. Add Book
2. Register Member
3. Borrow Book
4. Return Book
5. Display Books
6. Display Members
7. Search Book
8. Library Summary
9. Exit

Enter your choice: 2

Member ID: 212154

Member Name: Tushar

Member registered successfully.

```
===== LIBRARY MANAGEMENT SYSTEM =====
```

1. Add Book
2. Register Member
3. Borrow Book
4. Return Book
5. Display Books
6. Display Members
7. Search Book
8. Library Summary
9. Exit

Enter your choice: 3

Member ID: 212154

Book ID: 52648

Tushar borrowed 'Python' successfully.

```
===== LIBRARY MANAGEMENT SYSTEM =====
```

1. Add Book
2. Register Member
3. Borrow Book
4. Return Book
5. Display Books
6. Display Members
7. Search Book
8. Library Summary
9. Exit

Enter your choice: 4

Member ID: 212154

Book ID: 52648

Book returned successfully.

===== LIBRARY MANAGEMENT SYSTEM =====

1. Add Book
2. Register Member
3. Borrow Book
4. Return Book
5. Display Books
6. Display Members
7. Search Book
8. Library Summary
9. Exit

Enter your choice: 5

Library Books

=====

ID: 101
Title: Python Programming
Author: Guido Van Rossum
Status: Available

ID: 102
Title: Data Structures
Author: Mark Allen
Status: Available

ID: 103
Title: Machine Learning
Author: Andrew Ng
Status: Available

ID: 104
Title: Artificial Intelligence
Author: Stuart Russell
Status: Available

ID: 52648
Title: Python
Author: D.Arun Kumar
Status: Available

===== LIBRARY MANAGEMENT SYSTEM =====

1. Add Book
2. Register Member
3. Borrow Book
4. Return Book
5. Display Books
6. Display Members
7. Search Book
8. Library Summary
9. Exit

Enter your choice: 7

Enter book title: Python

Search Results

=====

ID: 101
Title: Python Programming
Author: Guido Van Rossum
Status: Available

ID: 52648
Title: Python
Author: D.Arun Kumar
Status: Available

===== LIBRARY MANAGEMENT SYSTEM =====

1. Add Book
2. Register Member
3. Borrow Book
4. Return Book
5. Display Books
6. Display Members
7. Search Book
8. Library Summary
9. Exit

Enter your choice: 7

Enter book title: Python

Search Results

=====

ID: 101

Title: Python Programming

Author: Guido Van Rossum

Status: Available

ID: 52648

Title: Python

Author: D.Arun Kumar

Status: Available

===== LIBRARY MANAGEMENT SYSTEM =====

1. Add Book
2. Register Member
3. Borrow Book
4. Return Book
5. Display Books
6. Display Members
7. Search Book
8. Library Summary
9. Exit

Enter your choice: 8

Library Summary

=====

Total Books : 5

Available Books : 5

Borrowed Books : 0

Total Members : 1

Conclusion:-

This project demonstrates how Object-Oriented Programming can be used to model a real-world library system. It reinforces core OOP concepts such as classes, objects, encapsulation, and object interaction while providing practical experience with Python data structures, control flow, and user interaction. It serves as a strong foundation for more advanced systems that include file handling, databases, due dates, fines, and graphical user interfaces.